CLASS 8 MATHEMATICS :: SLIP TEST – FA-3

Questions

Student Name: ..... **Roll No:** ..... **Class & Sec:** .....

Question 1.

Find the square root of 784 using the prime factorisation method.

Question 2.

Find the smallest natural number by which 256 must be multiplied so that the product becomes a perfect cube.

Question 3.

Find the square root of 12.25 using the division method.

Question 4.

The marks scored by 8 students in a mathematics test are 15, 22, 18, 25, 14, 20, 18, and 28. Find the range and the mean of this raw data.

Question 5.

For the class intervals 10–20, 20–30, and 30–40, determine the class size, lower limits, upper limits, and the class mark of the interval 20–30.

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Solutions

Solution 1:

$784 = 2 \times 2 \times 2 \times 2 \times 7 \times 7$

$\sqrt{784} = \sqrt{(2^2 \times 2^2 \times 7^2)}$

$= 2 \times 2 \times 7 =$ **28**

Solution 2:

$256 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$

$256 = (2^3) \times (2^3) \times (2^2)$

To complete the triplet, multiply by 2: $(2^2 \times 2 = 2^3)$

Smallest required natural number = **2**

Solution 3:

Pairing digits: 12 . 25

$3 \times 3 = 9$ | Remainder $= 12 - 9 = 3$

Divisor doubled: $3 \times 2 = 6$ ➔ $65 \times 5 = 325$

Remainder = 0 ➔ $\sqrt{12.25} =$ **3.5**

Solution 4:

Observations: 14, 15, 18, 18, 20, 22, 25, 28

Range = Maximum − Minimum = 28 − 14 = **14**

Mean = (14 + 15 + 18 + 18 + 20 + 22 + 25 + 28) / 8 = 160 / 8 = **20**

Solution 5:

Class size = 20 − 10 = **10**

Lower limits = **10, 20, 30** | Upper limits = **20, 30, 40**

Class mark of (20–30) = (20 + 30) / 2 = 50 / 2 = **25**